gnani-dub

A Chrome extension that dubs YouTube videos into Indian languages while you watch them.

THE PROBLEM

Most of the tutorials, lectures and explainers on YouTube are in English. Subtitles help, but only if you read the language and can read fast enough to keep up. For many people in India the easiest way to learn from a video is to hear it in their own language, and very few videos are dubbed.

THE SOLUTION

gnani-dub adds a **DUB** button to the YouTube player. Pick one of nine languages (Hindi, Bengali, Tamil, Telugu, Kannada, Malayalam, Marathi, Gujarati, Punjabi) and the video plays with a translated voice track and subtitles. There's nothing to upload and no waiting for a file to render. It works on the video you're already watching.

- **Videos with captions:** captions are translated ahead of playback and each line is spoken exactly when the video reaches it. If a line isn't ready, the video waits instead of drifting out of sync.
- **Videos without captions:** the extension listens. Audio streams to Gnani's speech-to-text, and the video pauses after each sentence while it's translated and spoken, like an interpreter taking turns.
- **The speaker's own voice (optional):** after about 15 seconds of speech, the dub switches to a clone of the original speaker's voice. It's off by default and meant for videos you own or have permission to use.

GNANI AI INSIDE

Model / API	What it does here
Timbre v3.0 (TTS)	Speaks every translated line
Prisma v3.0 streaming STT	Transcribes videos with no captions, live, with language detection; each sentence comes back 0.3 to 0.6 s after the speaker stops
Voice cloning (vachana-vc-v1)	Dubs in the original speaker's voice

Translation uses Sarvam. A small local server keeps API keys out of the browser and caches translations and clips, so rewatching or seeking back costs no credits.

WHY IT'S HARD, AND HOW IT'S HANDLED

- **Staying in sync:** translation runs about two minutes ahead and speech about 25 seconds ahead, and the first lines are prepared while the panel is still open, so the dub usually starts at once.
- **Getting the captions:** YouTube often blocks direct caption downloads. The extension falls back to the player's own copy, then the video's transcript, and finally to listening.
- **Streaming STT in a browser:** browsers can't send auth headers on a WebSocket, so audio goes through the local server. The server paces it at the 32 ms frames Gnani expects and starts a fresh session before the 15-minute limit.

WHO IT'S FOR AND WHAT'S NEXT

Students, first-time internet users and anyone learning from videos made in a language they don't speak. Creators can preview a dub of their own videos in their own voice. Next steps: an LLM pass for idioms and speaker gender, a voice per speaker, and support for Edge and Firefox.

Code: github.com/manit2004/gnani-dub · Built for the Great Indian AI Internship Challenge · #GreatIndianAIInternshipChallenge #GnaniAI